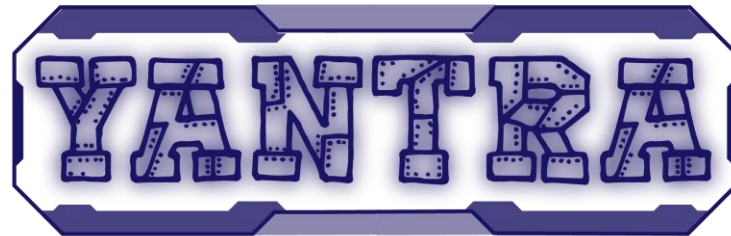


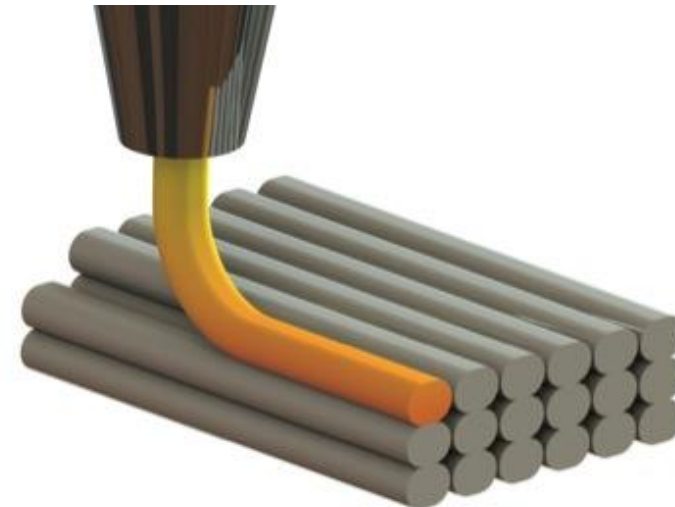
Fused deposition modeling (FDM)



Fused deposition modeling (FDM)

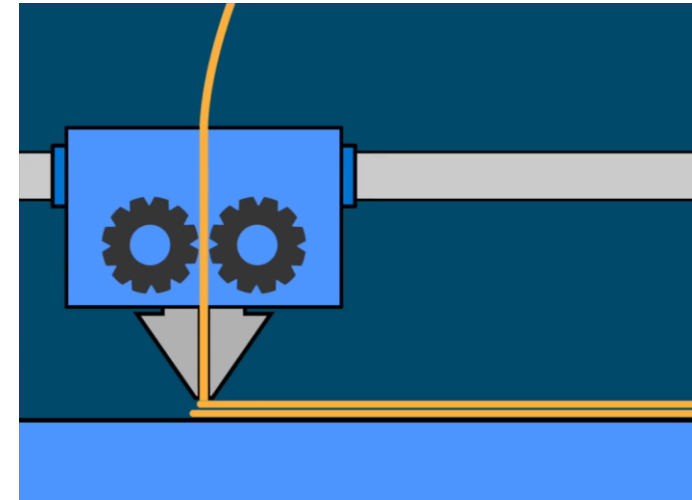
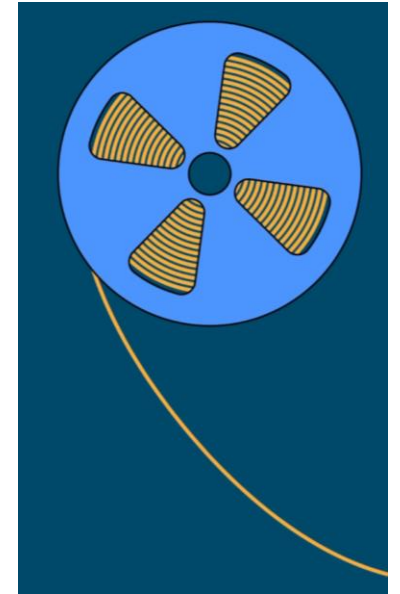
Remember glue gun while studying this topic

- The **second most widely** used RP technology, after stereolithography.
- The process is unlike all the other RP processes
- Because a **laser is not used** to produce the cross-sectional layer instead,
- The model material is applied directly in a semiliquid form from a CNC controlled extruder head.
- FDM is the favored type of RP process when **tough plastic components** are needed.



PRINCIPLE

- The principle of the FDM is based on surface chemistry, thermal energy, and layer manufacturing technology.
- The material in **filament (spool)** form is melted in a specially designed head, which extrudes on the model. As it is extruded, it is **cooled and thus solidifies** to form the model.
- The model is built **layer by layer**, like the other RP systems.
- Parameters which affect performance and functionalities of the system are material column strength, material flexural modulus, material viscosity, positioning accuracy, road widths, deposition speed, volumetric flow rate, tip diameter, envelope temperature, and part geometry.



Process

The process depends on building up the model layer-by-layer using a thermoplastic 0.003- mm-diameter filament.

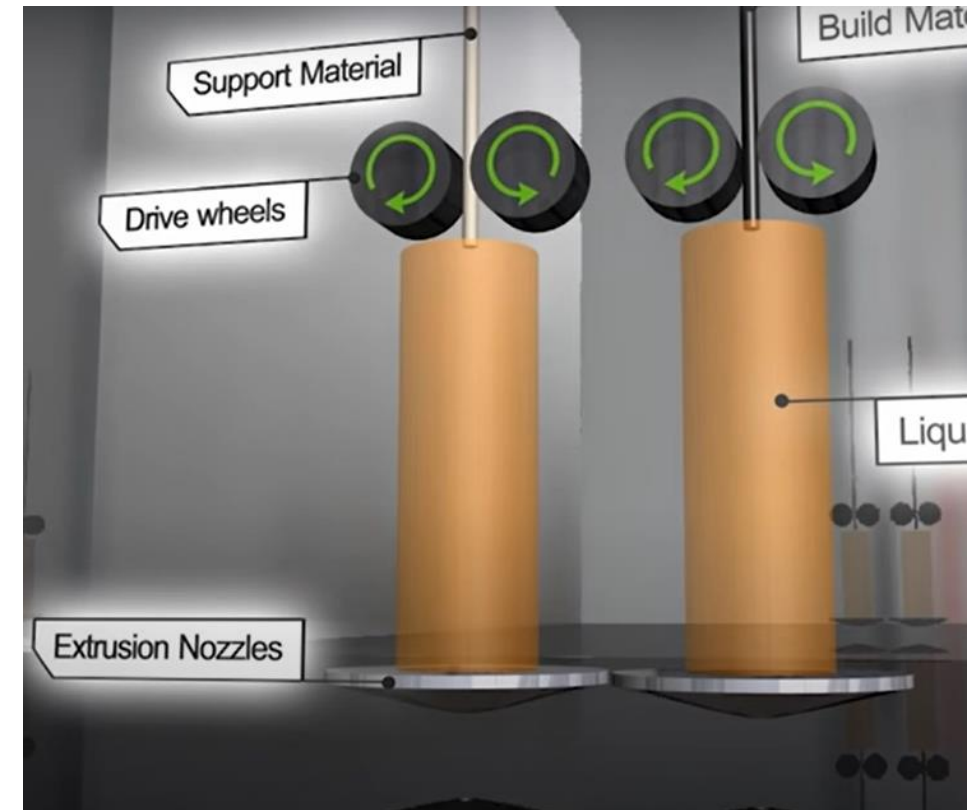
The filament is **unwound** from a coil and supplies material to an extrusion nozzle.

The filament **melts** as it flows through the heated extrusion head

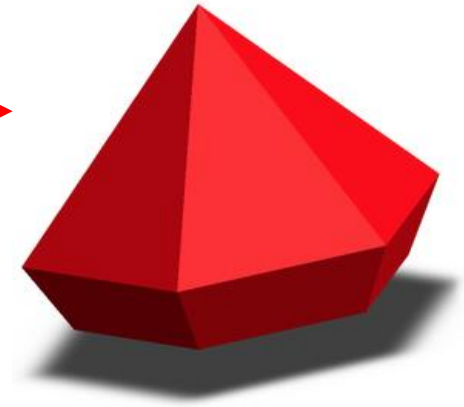
and emerges as a thin ribbon through an exit nozzle.



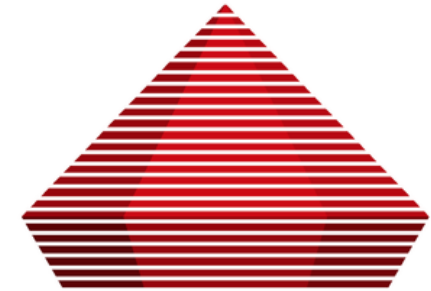
thermoplastic



1. The FDM system operating software imparts a **CAD** file.

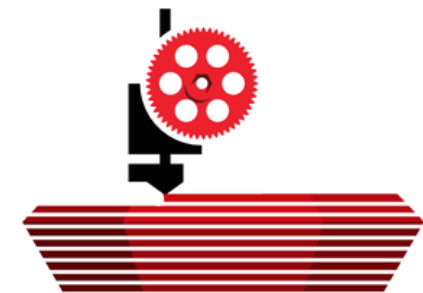


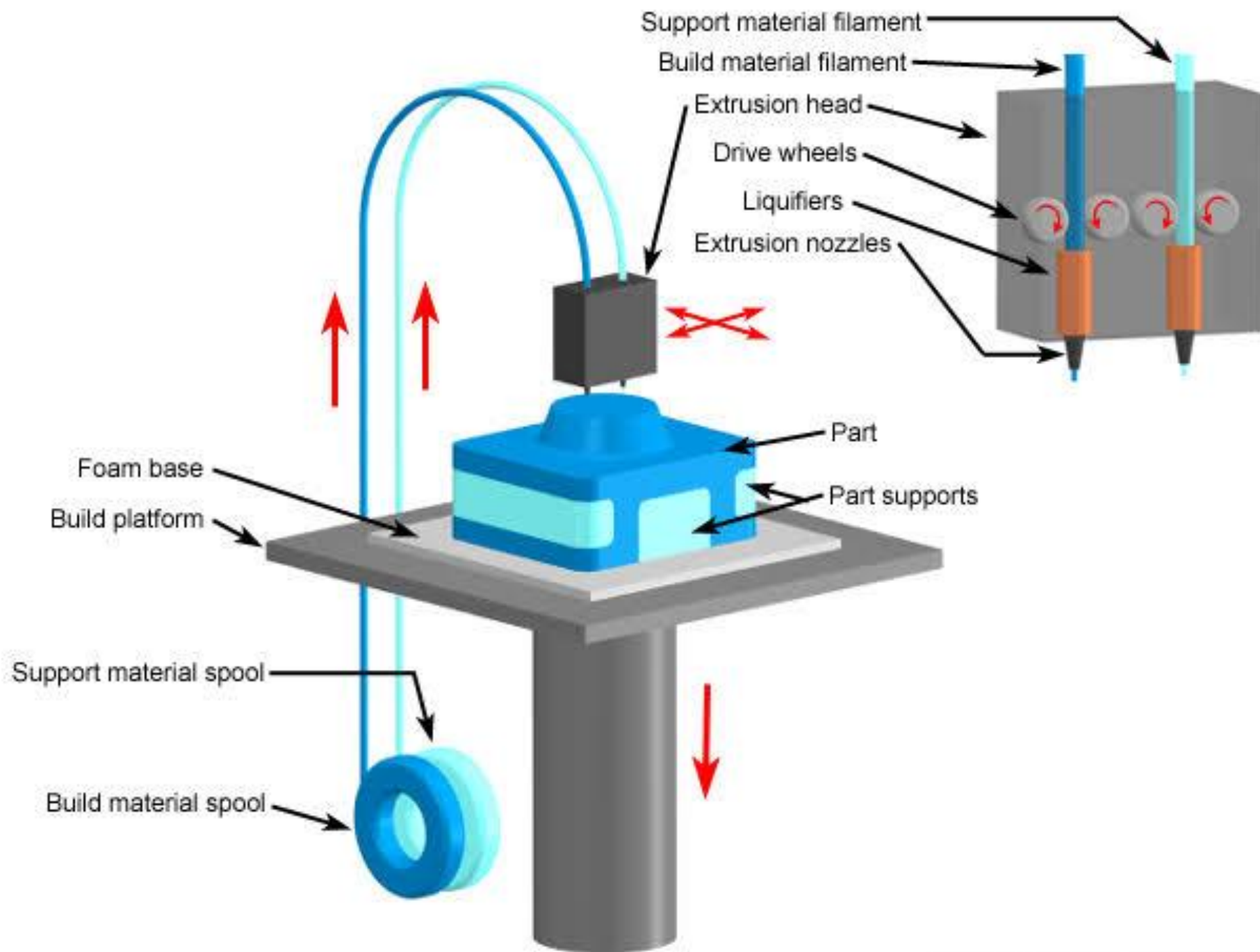
2. The software then orients the part for optimum building, **slices the model**,



3. and creates the path generated by the operating software, laying down a very thin layer of molten material.

4. One **layer upon another**, the model is built from the **bottom up**.



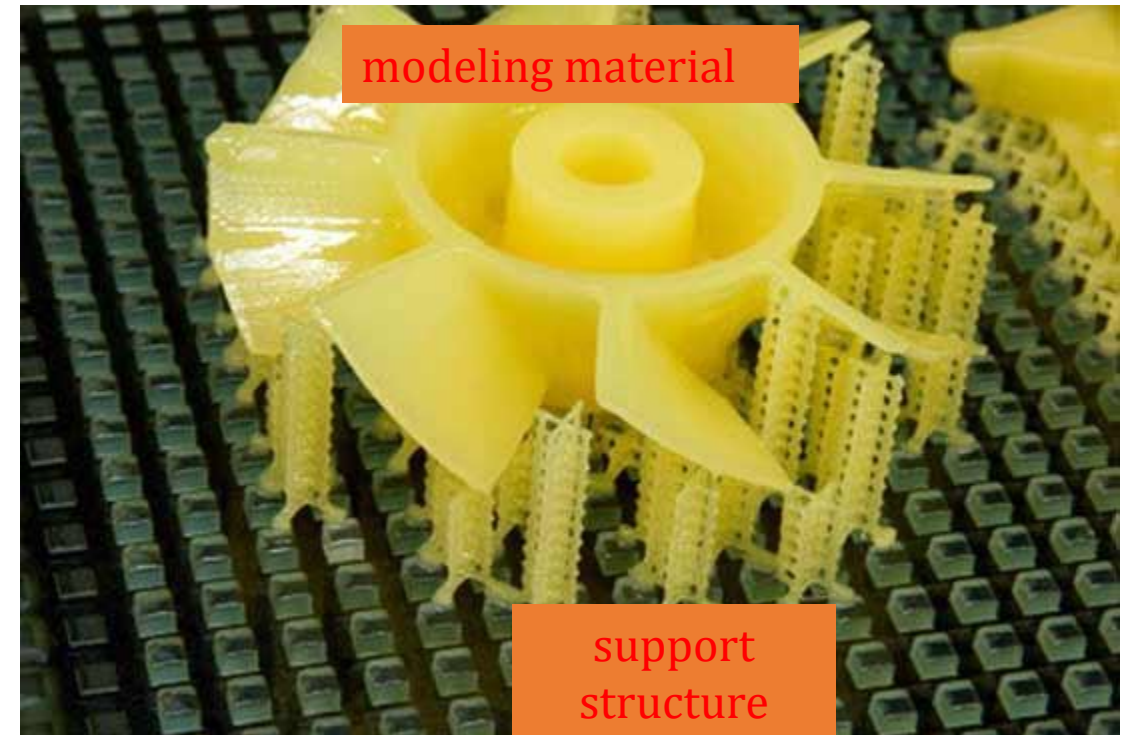


The required object is produced as a three-dimensional solid part
in a single stage without need for tooling.

Recent FDM systems have two extrusion heads:

one for modeling material

other for modeling support structure material

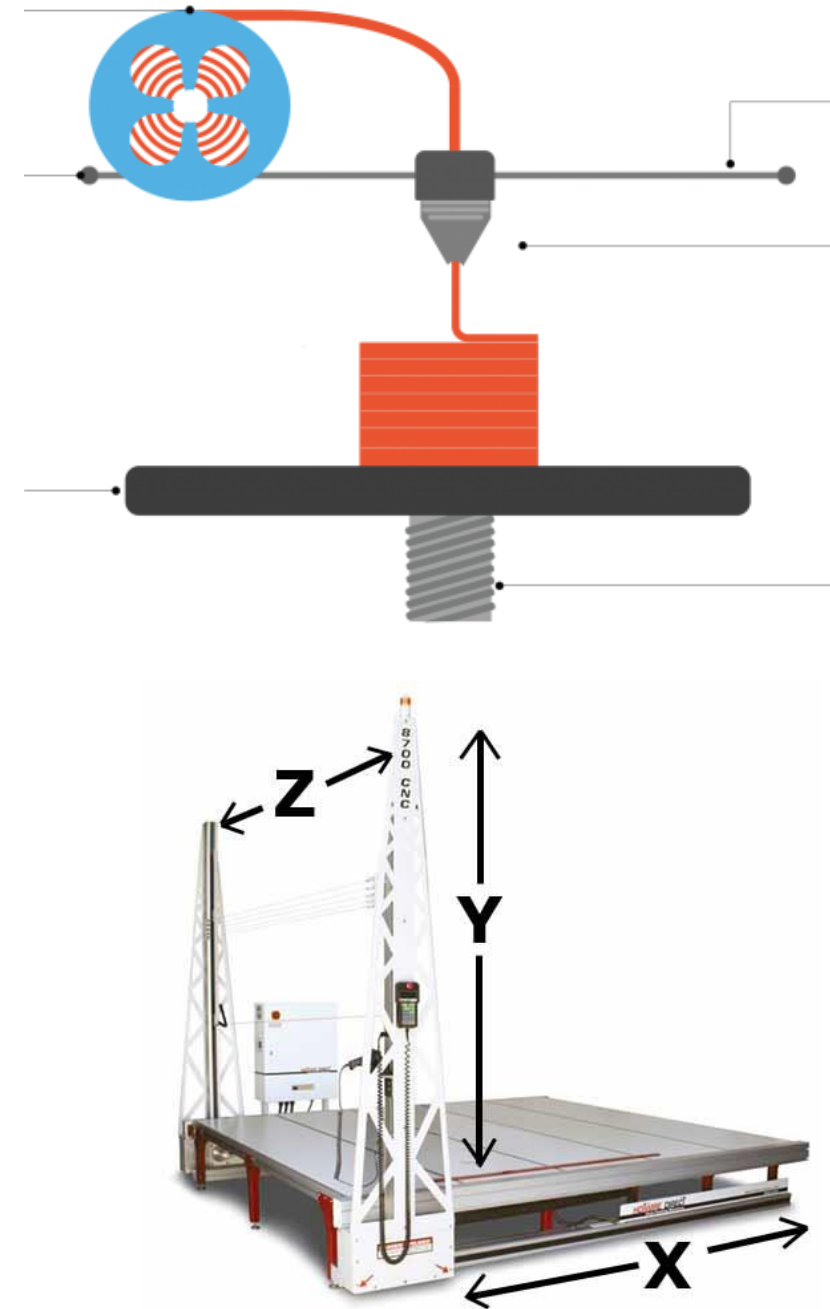


the **nozzle** is mounted to a mechanical stage, which can be guided around by **CNC code**

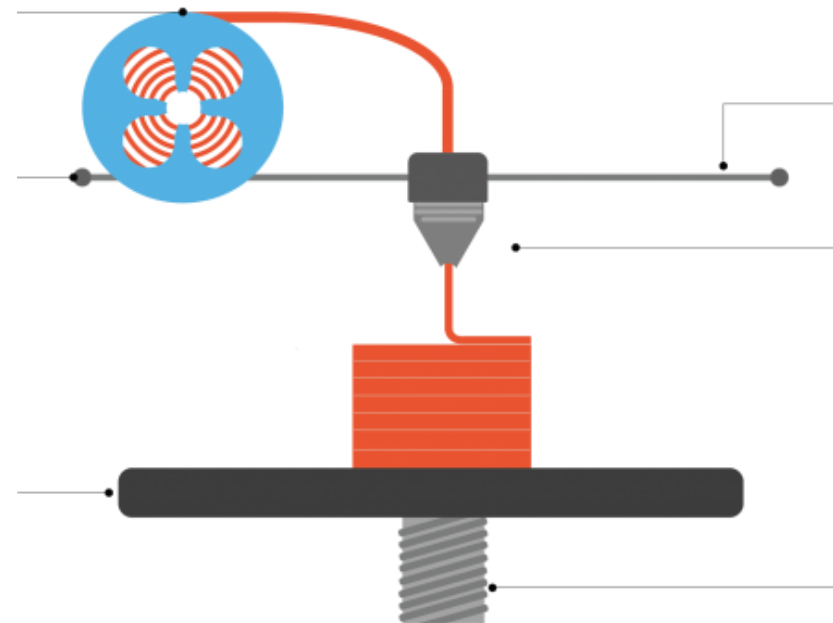
and the viscous ribbon of polymer is gradually built up from a fixtureless base plate

As the nozzle is moved over the table in the required **x, y, z geometry**, it deposits a thin bead of extruded plastic to form each layer.

The **plastic hardens** immediately after being squirted from the nozzle and bonds to the layer below.



- The entire system is **contained within a chamber**, which is held at a temperature just below the melting point of the plastic material.
- When the first layer is complete, the **nozzle** moves upward by one layer thickness and the process begins again.
- The part is easily removed from the platform,
- **supports** are **dissolved**, and the part becomes ready



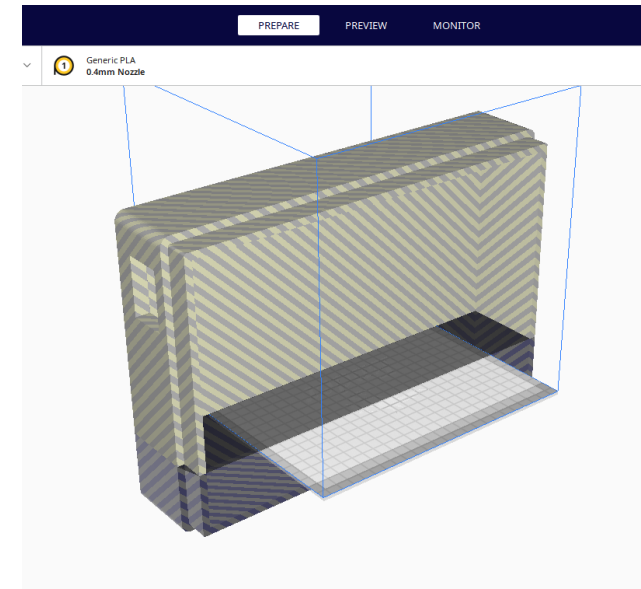
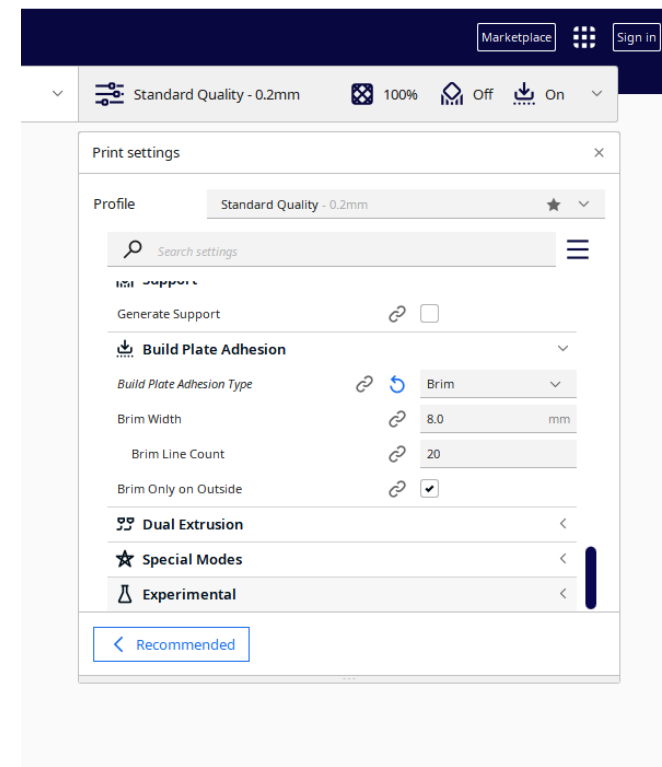
PATH GENERATION

1. CAD File Preparation:

- Before building a part, the STL file has to be converted into the machine language understood by the FDM.
- The STL file is read into slicing software, and is displayed graphically on screen in the Cartesian coordinate system (x, y and z).
- Slicing software gives you options on the FDM system being used, the slice layer thickness, the build and support materials, as well as the tip sizes.

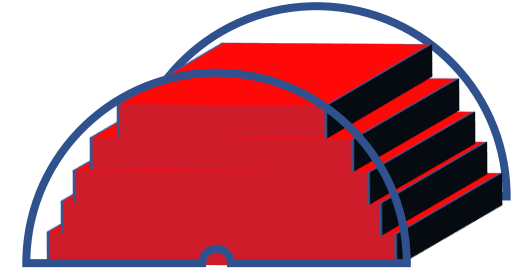
2. Part Size:

- First it must be affirmed that the part will fit into the bounding box;
- In some cases, for instance if the part fits in x and y but is too tall in the z, Slicing software can be used to section the part by slicing to a certain height, then starting a new build later at that height and finishing the part.
- This technique results in flat mating surfaces with no alignment bosses or slots, therefore it is up to the post-processing person to align the subsections properly during bonding.



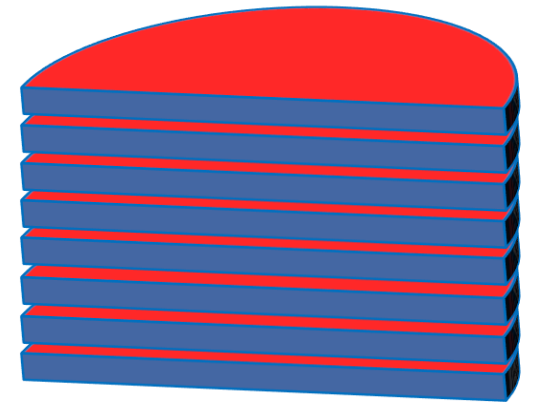
3. Orientation/Positioning:

- Once the part (or parts) has been deemed an appropriate build size, the part should be oriented in an optimum position for building.
- Also, rounded surfaces tend to turn out smoother if built in the plane of movement of the extrusion head (x, y), as curvatures in the z direction are affected by the layering build technique.



4. Slicing:

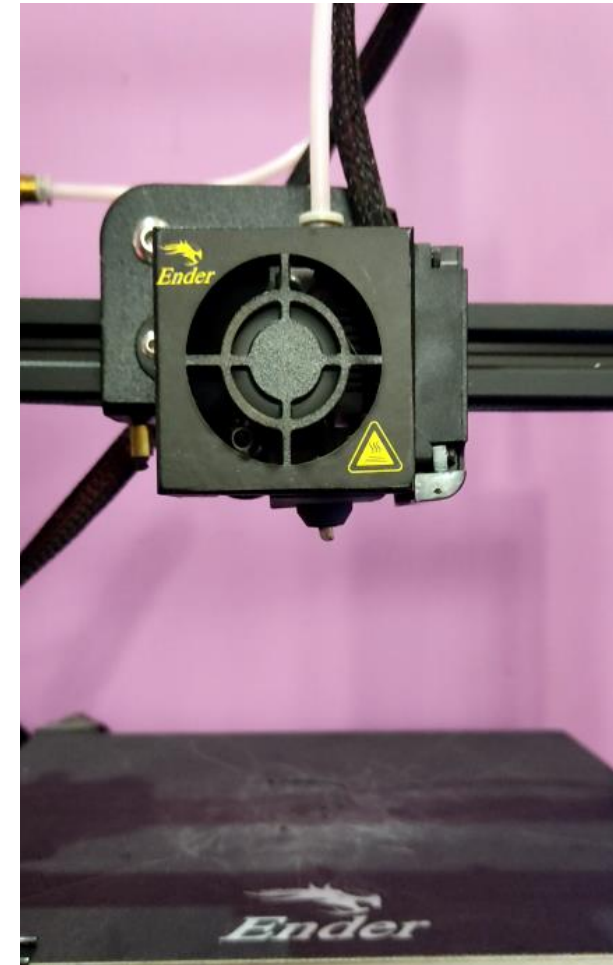
- Once the part is been properly oriented and scaled, it must be sliced. Slicing is a software operation that creates thin, horizontal cross sections of the STL file that will later be used to create the control code for the machine.
- Thinner slices may be used for higher definition models, but this increases the time required to complete a part build. Likewise, less accuracy-sensitive parts can be built much faster using a thicker slice value.
- There becomes a tradeoff between the desired accuracy and the time needed, with the optimum value determined by the user.



Process Parameters

1. Liquifier temperature:

- The heating element is electronically controlled, and has **feedback thermocouples** to allow for a stable temperature throughout.
- The heating elements are held at a temperature just **above the melting point** of the material, so that the filament passing from the exit of the chamber is in a **semi-molten state**.
- This allows for smooth extrusion as well as tight control on the material placement.



2. Nozzle diameter:

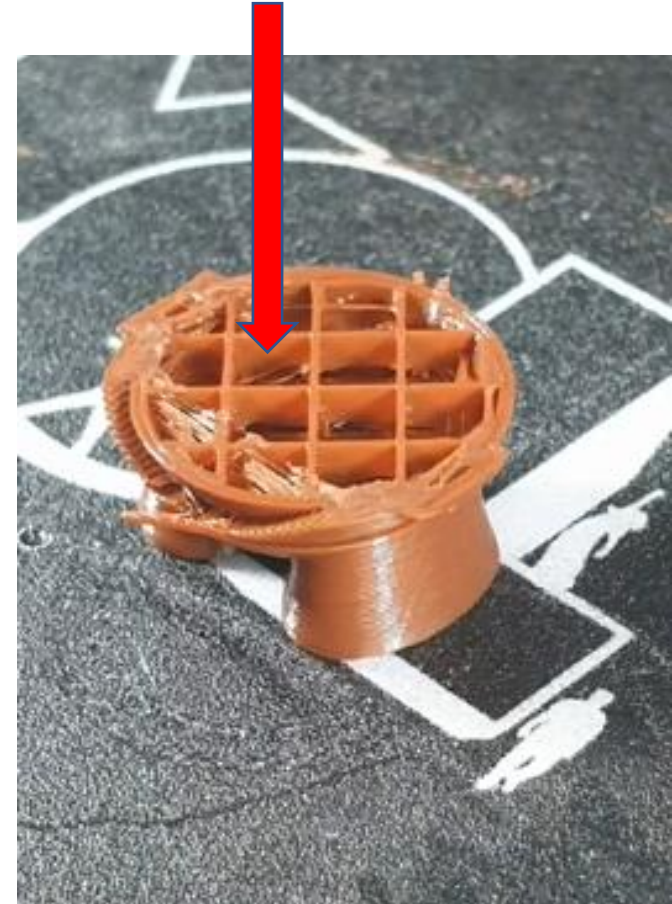
- The nozzles are externally threaded and screw up into the heating chamber exit, and are used to reduce the extruded filament diameter to allow for better detailed modeling.
- The tips are heated by the heating chamber up to above the melting point of the material.
- The tips can be removed and replaced with different size openings, the two most common being the 0.012 and 0.025 inch sizes.



3. Road width:

- The road width is the width of the ribbon of molten material that is extruded from the tip.
- When the FDM builds a layer, it usually begins by outlining the cross section with a perimeter road, sometimes followed by one or more concentric contours inside of the perimeter. Next, it begins to fill the remaining internal area in a raster, or hatch, pattern until a complete solid layer is finished.
- The road width can be as small as the diameter of the tip opening to approximately twice that size.
- The width is controlled by increasing or decreasing the extrusion rate in conjunction with the speed of the head

- 4. Fill spacing:
- Fill spacing is the distance left between the rasters (or) contours that make up the interior solids of the part.
- A fill spacing set at zero just means that the part will be built solid.
- a bulky part can be significantly reduced in time and material cost by leaving air gaps between each consecutive road.
- Keep in mind that to make sure the part appears solid, all exposed surfaces need to be built with zero fill, or else you will be able to see the open spacing.



Advantages of FDM

- Quiet, **nontoxic**, and office-friendly process
- **Fast** for small, thin shapes
- Produces **strong** parts
- **Cost-effective**
- No need for part cleanup
- Able to produce **multicolored** parts using colored ABS



multicolored parts

Disadvantages of FDM

- **Slow** when making large cross-sectional areas
- Has to use **additional support material**
- Relatively **low accuracy and poor surface finish**
- Unable to make **complicated parts**
- Limited number of materials available

ADVANTAGES

1. Fabrication of functional parts:

FDM process is able to fabricate **prototypes** with materials that are similar to that of the actual molded product. With ABS, it is able to fabricate **fully functional parts** that have 85% of the strength of the actual molded part. This is especially useful in developing products that require quick prototypes for functional testing.

2. Minimal wastage:

The FDM process build parts directly by extruding semi-liquid melt onto the model. Thus only those material needed to build the part and its support are needed, and material wastages are kept to a minimum. There is also little need for cleaning up the model after it has been built.

3. Ease of support removal: With the use of Break Away Support System (BASS) and Water Works Soluble Support System, support structures generated during the FDM building process can be easily broken off or simply washed away. This makes it very convenient for users to get to their prototypes very quickly and there is very little or no post-processing necessary.

4. Ease of material change: Build materials, supplied in spool form (or cartridge form in the case of the Dimension or Prodigy Plus), are easy to handle and can be changed readily when the materials in the system are running low. This keeps the operation of the machine simple and the maintenance relatively easy.

Disadvantages

- **Restricted accuracy:** Parts built with the FDM process usually have restricted accuracy due to the shape of the material used, i.e., the filament form. Typically, the filament used has a diameter of 1.27 mm and this tends to set a limit on how accurately the part can be built.
- **Slow process:** The building process is slow, as the whole cross-sectional area needs to be filled. Building speed is restricted by the extrusion rate or the flow rate of the build material from the extrusion head. As the build material used are plastics and their viscosities are relatively high, the build process cannot be easily speeded up.

Disadvantages

3. **Unpredictable shrinkage**: As the FDM process extrudes the build material from its extrusion head and cools them rapidly on deposition, **stresses** induced by such rapid cooling invariably are introduced into the model.

As such, shrinkages and distortions caused to the model built are a common occurrence and are usually difficult to predict, though with experience, users may be able to compensate for these by adjusting the process parameters of the machine.

Applications

Models for conceptualization and presentation:

Models can be marked, sanded, painted and drilled and thus can be finished to be almost like the actual product.

Prototypes for design, analysis and functional testing:

The system can produce a fully functional prototype in ABS. The resulting ABS parts have 85% of the strength of the actual molded part. Thus actual testing can be carried out, especially with consumer products.

Patterns and masters for tooling:

Models can be used as patterns for investment casting, sand casting and molding.

Medical Applications:

The Medical Grade ABS has been approved by the U.S. Food and Drug Administration (FDA), and therefore is used by the medical industry to produce various parts within the industry. Since CAT Scan and MRI data can be converted into the .STL file format, custom models of internal organs, bones, etc. can be reconstructed and studied before a patient ever goes into surgery.

Flexible Components:

The recently released Elastomer material opens yet another dimension of functionality for the FDM systems. Flexible test components such as seals, shrouds, and tubing can be prototyped with the Elastomer material to proof out the concepts or "make the sale" on crucial designs.